M E I T Y A I C Y B E R P R O J E C T

Objective 1: Standardisation of Cybersecurity Audit Report Format

DEMO VISUALISATION MANUAL

AI-Assisted Cybersecurity Audit Report Standardisation System

End-to-End Demonstration across VAPT • Network • Web Application • IoT Domains

1. Purpose and How to Read This Manual

1.1 Purpose

This Demo Visualisation Manual provides a step-by-step illustrated walkthrough of the AI-Assisted Cybersecurity Audit Report Standardisation System. It demonstrates how the system processes a real cybersecurity audit report — from raw, unstructured upload to a clean, machine-readable, source-traceable standardised JSON output — across all four audit domains covered by Objective 1: VAPT, Network Infrastructure, Web Application Security, and IoT Security. The manual is designed to support evaluation, demonstration to stakeholders, and explanation to non-technical reviewers. Every stage of the 13-stage processing pipeline is shown with its input, the processing action performed, and the output produced. Actual sample inputs and sample structured outputs are shown side-by-side for each domain.

1.2 What This Manual Demonstrates

Section	What is Demonstrated
Section 2	The complete 13-stage pipeline shown as a visual flow with stage-by-stage input → process → output
Section 3	Live VAPT demo: unstructured PDF report → structured JSON output, with all intermediate stages shown
Section 4	Live Network Infrastructure Audit demo: full pipeline walkthrough
Section 5	Live Web Application Security demo: full pipeline walkthrough
Section 6	Live IoT Security Audit demo: full pipeline walkthrough
Section 7	Demonstration of source traceability: how every extracted field links back to its original location
Section 8	Demonstration of AI synonym mapping: how different headings are mapped to standard sections
Section 9	Demonstration of severity normalisation: raw labels → standard 5-class output
Section 10	Final standardised output: JSON, XML, and PDF summary formats shown
Section 11	Evaluation metrics: confidence scores, processing times, domain coverage

1.3 Notation Used in This Manual

Symbol / Style	Meaning
INPUT BOX (blue border, left)	Raw content from the original uploaded audit report — exactly as found.
STAGE BANNER (teal background)	A numbered pipeline stage being demonstrated.
OUTPUT PANEL (navy header)	The structured output produced by that stage.
FINDING CARD (severity-coloured border)	One extracted, normalised vulnerability finding.
Dark code block	JSON, XML, or code sample showing exact system output.
Confidence %	AI extraction confidence: >= 85% High (green), 60–84% Medium (amber), < 60% Low (red).

2. System Pipeline — Visual Overview

2.1 The 13-Stage Processing Pipeline

Every audit report — regardless of format, domain, or originating agency — passes through the same 13 stages in sequence. Each stage is independent and produces a well-defined output consumed by the next stage.

01 **SECURE UPLOAD — User uploads audit report file through the system interface**

02 **FILE VALIDATION — MIME-type check, size check, password-protection check**

03 **FILE HASHING — SHA-256 hash computed for integrity and traceability**

04 **MALWARE SCAN — ClamAV offline scan; FLAGGED files are quarantined immediately**

05 **OCR PROCESSING — Tesseract v5 + OpenCV applied to scanned/image-based pages only**

06 **TEXT & TABLE EXTRACTION — Native text extracted; tables extracted via camelot-py**

07 **EVIDENCE EXTRACTION — Screenshots, logs, tool outputs extracted and hashed**

08 **AI SECTION DETECTION — ModernBERT + BGE-M3 classify headings to standard sections**

09 **RULE-BASED PARSING — 27 regex rules extract CVE, CVSS, IP, severity, URL, etc.**

10 **AI ENTITY EXTRACTION — Mistral-7B extracts structured findings from section text**

11 **SEVERITY NORMALISATION — SetFit normalises raw labels to 5-class standard**

12 **SOURCE TRACEABILITY — Every field tagged with file, page, paragraph, snippet**

13 **SCHEMA MAPPING & OUTPUT — JSON / XML / PDF structured outputs generated**

2.2 Pipeline Input/Output Summary

Stage	Key Input	Key Output	Time (typical)
01 Upload	User-selected file	File stored in secure temp dir	< 1 sec
02 Validation	File path	Format, size, MIME confirmed	< 1 sec
03 Hashing	File bytes	SHA-256 hex string	< 1 sec
04 Malware Scan	File path	CLEAN / FLAGGED status	2–15 sec
05 OCR	PDF/image pages	Text per page + confidence score	5–30 sec (scanned docs)
06 Text/Table	Document file	raw_pages list, tables list	2–10 sec
07 Evidence	Document file	Evidence file list with hashes	1–5 sec
08 Section Detection	raw_pages	Sections with standard names + confidence	10–30 sec
09 Rule Parsing	Full text	27-category entity dict	< 1 sec
10 Entity Extraction	Sections + tables	Findings, tools, summaries	30–120 sec (LLM)
11 Severity Norm.	Findings with raw severity	Findings with normalised severity	< 2 sec
12 Traceability	Entities + raw pages	Entities with source_traceability	2–5 sec
13 Output Gen.	Full schema dict	JSON + XML + PDF files	2–5 sec

3. Domain Demo 1 — VAPT (Vulnerability Assessment and Penetration Testing)

3.1 Sample Input Report — Description

> **Sample Input File**
> Filename: vapt_report_agency_abc_sep2026.pdf | Pages: 28 | Format: Text-based PDF | Agency: XYZ Cyber Security Pvt. Ltd. | Submitted: 18 Sep 2026 | Audit Type: External Grey-Box VAPT

The sample VAPT report submitted by the agency has the following structure — note how it differs from the standard schema. This is exactly the kind of heterogeneity the system resolves:

Agency Section Heading (Original)	Standard Section Name (After Mapping)	Mapping Method	Confidence
Project Overview	Report Metadata + Audited Organisation Details	Synonym Rule	95%
Engagement Scope	Audit Scope	Synonym Rule (Exact)	100%
Assessment Methodology	Audit Methodology	ModernBERT Classifier	88%
Target Systems	Assets Covered	BGE-M3 Similarity	79%

Observations	Vulnerability Findings	Synonym Rule (Partial)	92%
Corrective Measures	Recommendations	Synonym Rule (Exact)	100%
Management Summary	Executive Summary	Synonym Rule	93%
Technical Results	Technical Summary	ModernBERT Classifier	85%
Closing Remarks	Conclusion	Synonym Rule (Partial)	91%

3.2 Stage-by-Stage VAPT Demo

STAGE 05 **OCR Processing — VAPT Report**

This VAPT report is a native text-based PDF. OCR is NOT applied to text pages. PyMuPDF extracts native text directly from 27 of 28 pages. Page 14 contains a scanned screenshot evidence attachment — OCR is applied only to this page.

OCR Stage Output	
OCR Applied?	False (27 native pages) + True (Page 14 — scanned screenshot)
Pages Processed	28
Avg. Confidence	0.97 (native) / 0.83 (scanned page 14)
Processing Time	1.8 seconds

STAGE 06 **Text & Table Extraction — VAPT Report**

PyMuPDF extracts 28 pages of text. camelot-py extracts 3 tables from the findings section (pages 8–16): one findings summary table, one CVSS score table, and one remediation priority table.

Table Extraction Output	
Tables Found	3 tables
Table 1 (Page 8)	Findings Summary — columns: Finding No., Title, Severity, CVSS Score, Page Ref.
Table 2 (Pages 9–16)	Detailed Findings — columns: ID, Vulnerability, Affected System, Severity, CWE, Recommendation
Table 3 (Page 22)	Remediation Priority Matrix — Priority, Finding ID, Estimated Effort

STAGE 08 **AI Section Detection — VAPT Report**

ModernBERT processes all 9 detected headings. 6 are matched via synonym rules, 2 via ModernBERT, 1 via BGE-M3 similarity. All 9 headings are successfully mapped to standard section names.

Original Heading Found	**→ Standard Section**	**Method**	**Conf.**
"Observations" (Page 8)	Vulnerability Findings	Synonym Rule	92%
"Corrective Measures" (Page 17)	Recommendations	Synonym Rule	100%
"Target Systems" (Page 3)	Assets Covered	BGE-M3	79%
"Assessment Methodology" (Page 4)	Audit Methodology	ModernBERT	88%

"Management Summary" (Page 23)	Executive Summary	Synonym Rule	93%

STAGE 09 Rule-Based Parsing — VAPT Report

The rule parser scans all 28 pages in under 1 second and extracts the following entities:

Rule Name	Matches Found	Sample Values
CVE_ID	5	CVE-2024-21413, CVE-2023-44487, CVE-2024-1709, CVE-2023-4966, CVE-2024-3400
CVSS_SCORE	5	9.8, 8.8, 7.5, 6.5, 5.3
CVSS_VECTOR	3	CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H
CWE_ID	4	CWE-89, CWE-79, CWE-287, CWE-22
IP_V4	12	192.168.1.100, 192.168.1.101, 10.0.0.5, 172.16.0.1 ...
URL	7	https://target.gov.in/admin, /api/v1/login, /portal/upload ...
SEVERITY_LABEL	18	critical (x2), high (x7), medium (x6), low (x3)
HTTP_METHOD	9	POST (x5), GET (x3), PUT (x1)
OWASP_CATEGORY	4	A01:2021, A03:2021, A07:2021, A10:2021

STAGE 10 AI Entity Extraction — Mistral-7B — VAPT Report

Mistral-7B-Instruct processes the Vulnerability Findings section (pages 8–16, ~4,200 words). It extracts 12 structured finding records. Below are 3 representative finding cards showing the extracted output:

FIND-001 SQL Injection in Login API [Critical]

CVSS: 9.8 **Asset:** 192.168.1.100 (/api/v1/login) **CVE:** CVE-2024-21413 **CWE:** CWE-89

Recommendation: Use parameterised queries / prepared statements. Input validation must be enforced server-side.

Confidence: 94% Source: Page 9

FIND-002 Stored XSS in User Profile Module [High]

CVSS: 8.2 **Asset:** https://target.gov.in/profile **CVE:** CVE-2023-44487 **CWE:** CWE-79

Recommendation: Encode all user-supplied output. Implement Content Security Policy (CSP) headers.

Confidence: 88% Source: Page 11

FIND-004 Insecure Direct Object Reference (IDOR) — Document Download [High]

CVSS: 7.5 **Asset:** https://target.gov.in/api/docs/{id} **CVE:** — **CWE:** CWE-22

Recommendation: Implement authorisation checks on all object references. Validate user permissions server-side.

Confidence: 85% Source: Page 13

STAGE 11 **Severity Normalisation — VAPT Report**

The VAPT report used mixed severity labels: 'P0' (2 findings), 'Major' (3 findings), 'Moderate' (4 findings), 'Low Risk' (2 findings), 'Info' (1 finding). SetFit normalises all to the 5-class standard:

Raw Label in Report	Normalised Standard Label	Method	Confidence
P0	Critical	SetFit Classifier	99%
Major	High	SetFit Classifier	95%
Moderate	Medium	Synonym Rule	100%
Low Risk	Low	Synonym Rule	100%
Info	Informational	Synonym Rule (Exact)	100%

STAGE 13 **Final Output — VAPT Report**

The complete standardised output for the VAPT report. Summary statistics:

VAPT Structured Output Summary	
Submission Reference	SRN-2026-VAPT-A3F2C1D4
Total Findings	12
Critical	2 (FIND-001, FIND-003)
High	5 (FIND-002, FIND-004, FIND-006, FIND-008, FIND-010)
Medium	3 (FIND-005, FIND-007, FIND-009)
Low / Info	2 (FIND-011, FIND-012)
Overall Risk Rating	Critical
CVEs Mapped	5 of 12 findings
CWEs Mapped	10 of 12 findings
Compliance Mapped	OWASP Top 10 2021 (all), ISO 27001:2022 (all)
Evidence Items	8 screenshots, 2 tool output logs
Traceability	All 12 findings source-traced to page + paragraph
Avg. Confidence	91%
Total Processing Time	~68 seconds
Output Formats	JSON + XML + PDF Summary + ZIP Package

4. Domain Demo 2 — Network Infrastructure Audit

4.1 Sample Input Report — Description

> **Sample Input File**
> Filename: network_audit_ministry_oct2026.docx | Pages: 34 | Format: DOCX (Word Document) | Agency: SecureNet Auditors | Audit Type: Internal Network Infrastructure Audit

This DOCX report uses a non-standard structure with section headings like 'Network Assessment Findings', 'Configuration Issues', 'Open Ports Analysis', and 'Hardening Recommendations'. The system maps all of these to standard schema sections.

4.2 Key Processing Stages — Network Audit

STAGE 06 Text & Table Extraction — Network Audit (DOCX)

python-docx extracts 34 pages of text and 6 native Word tables. Key tables extracted include: Open Ports Summary (12 rows), Firewall Rule Issues (8 rows), Vulnerability Findings (15 rows), CIS Controls Compliance (18 rows).

STAGE 09 Rule-Based Parsing — Network Audit

Rule Name	Matches Found	Sample Extracted Values
IP_V4	47	192.168.0.0/24 range, 10.10.0.x range — 47 unique host IPs
PORT_NUMBER	23	22 (SSH), 23 (Telnet), 80, 443, 3389 (RDP), 8080, 5900 (VNC)...
CVE_ID	8	CVE-2023-20269, CVE-2024-20353, CVE-2023-27997 ...
SEVERITY_LABEL	22	critical (x3), high (x8), medium (x7), low (x4)
IP_V6	3	fe80::1, 2001:db8::1 — IPv6 found on 3 network devices
MAC_ADDRESS	5	00:1A:2B:3C:4D:5E — 5 device MACs extracted from scan output

STAGE 10 AI Entity Extraction — Network Findings

Three representative network findings extracted by Mistral-7B:

FIND-001 Telnet Service Running on Network Devices [Critical]

CVSS: 9.1 **Asset:** 192.168.1.1 (Core Router), Port 23 **CVE:** CVE-2023-20269 **CWE:** CWE-319

Recommendation: Disable Telnet immediately. Replace with SSH v2 with key-based authentication only.

Confidence: 96% Source: Page 9

FIND-003 RDP Exposed Externally Without MFA [High]

CVSS: 8.8 **Asset:** 192.168.0.50 (Windows Server 2019), Port 3389 **CVE:** CVE-2024-20353 **CWE:** CWE-308

Recommendation: Restrict RDP to VPN only. Enforce MFA. Apply latest Windows Server patches.

Confidence: 89% Source: Page 13

FIND-007 Default SNMP Community String 'public' [High]

CVSS: 7.8 **Asset:** 192.168.1.0/24 — 6 network devices affected **CVE:** CVE-2023-27997 **CWE:** CWE-521

Recommendation: Change SNMP community strings to complex values. Use SNMPv3 with authentication.

Confidence: 91% Source: Page 18

4.3 Network Audit — Domain-Specific Output Fields

Because this is a Network Infrastructure Audit, the system additionally populates the network_specific top-level object in the structured output. This object is only present for network domain reports:


```
"network_specific": {
  "network_topology_description": "Flat network with single core router, no segmentation.",
  "ip_ranges_scanned": ["192.168.0.0/24", "10.10.0.0/16"],
  "total_hosts_discovered": 47,
  "total_open_ports": 312,
  "firewall_present": true,
  "ids_ips_present": false,
  "network_segmentation_present": false,
  "dmz_present": false,
  "vpn_present": true,
  "wireless_networks_found": 2,
  "open_services_summary": [
    { "service": "Telnet", "port": 23, "host_count": 4, "risk_level": "Critical" },
    { "service": "RDP",    "port": 3389, "host_count": 2, "risk_level": "High"    },
    { "service": "VNC",    "port": 5900, "host_count": 1, "risk_level": "High"    }
  ],
  "cis_controls_coverage": [
    { "control_id": "CIS-01", "control_name": "Inventory of Enterprise Assets", "status":
"Partially Compliant" },
    { "control_id": "CIS-04", "control_name": "Secure Config of Enterprise Assets", "status":
"Non-Compliant"      },
    { "control_id": "CIS-12", "control_name": "Network Infrastructure Management",  "status":
"Non-Compliant"      }
  ]
}
```


Network Audit Output Summary	
Submission Reference	SRN-2026-NET-B7E4A2F1
Total Findings	15
Critical	3 | High: 8 | Medium: 3 | Low: 1
Overall Risk Rating	Critical
CVEs Mapped	8 of 15 findings
CIS Controls Coverage	18 controls tested — 5 Compliant, 8 Partial, 5 Non-Compliant
Hosts in Scope	47 hosts across 2 IP ranges
Traceability	All 15 findings traced to source page and table row
Avg. Confidence	89%

Processing Time	~72 seconds (DOCX — no OCR needed)

5. Domain Demo 3 — Web Application Security Testing

5.1 Sample Input Report — Description

Sample Input File
Filename: webapp_security_egovportal_aug2026.pdf | Pages: 42 | Format: PDF (Text-based) | Application: e-Governance Citizen Services Portal | Agency: AppSec India LLP | Audit Type: Black-Box Web Application Security Testing

This web application security report uses OWASP Top 10 2021 as its primary classification framework. Section headings include 'Vulnerability Details', 'Security Observations', 'Risk Classification', and 'Fixes and Mitigations'. The report also contains embedded HTTP request/response screenshots as evidence.

5.2 Key Processing Stages — Web Application

STAGE 09 **Rule-Based Parsing — Web Application Report**

Rule Name	Matches	Sample Values
URL	31	https://egovportal.gov.in/login, /api/citizens/{id}/docs, /admin/upload
HTTP_METHOD	24	POST (x11), GET (x9), PUT (x3), DELETE (x1)
HTTP_STATUS	19	200 (x8), 302 (x3), 403 (x2), 500 (x4), 401 (x2)
OWASP_CATEGORY	10	A01:2021 (x3), A02:2021 (x2), A03:2021 (x3), A07:2021 (x2)
CVE_ID	4	CVE-2024-1709, CVE-2023-4966, CVE-2024-3400, CVE-2023-50164
CVSS_SCORE	8	9.4, 8.8, 8.5, 7.5, 6.9, 6.5, 5.8, 4.3

STAGE 07 **Evidence Extraction — Web Application Screenshots**

The web application report contains 16 embedded screenshots showing HTTP requests, server responses, and tool outputs. The evidence handler extracts all 16, computes their SHA-256 hashes, and registers them as evidence items linked to their respective findings.

Evidence Extraction Output	
Total Evidence Items	16 screenshots + 3 tool outputs (Burp Suite exports)
Evidence Types	HTTP Request/Response (11), Tool Output — Burp (3), Error Page Screenshot (2)
Linked to Findings	14 of 16 evidence items successfully linked to a specific finding
Unlinked Items	2 items (general tool screenshots) stored as orphan evidence

STAGE 10 **AI Entity Extraction — Web Application Findings**

Four representative web application findings:

FIND-001 Broken Access Control — Admin Panel Exposed [Critical]

CVSS: 9.4 **Asset:** https://egovportal.gov.in/admin **CVE:** CVE-2024-1709 **CWE:** CWE-284

Recommendation: Implement role-based access control. Restrict /admin path to authorised IPs only. Enable MFA.

Confidence: 95% Source: Page 8

FIND-003 SQL Injection — Citizen Search API [Critical]

CVSS: 9.1 **Asset:** https://egovportal.gov.in/api/v1/search?q= **CVE:** CVE-2024-3400 **CWE:** CWE-89

Recommendation: Use ORM with parameterised queries. Disable detailed SQL error messages in production.

Confidence: 92% Source: Page 14

FIND-005 Reflected XSS — Feedback Form [High]

CVSS: 7.5 **Asset:** https://egovportal.gov.in/feedback **CVE:** — **CWE:** CWE-79

Recommendation: Encode all reflected user input. Set X-XSS-Protection and Content-Security-Policy headers.

Confidence: 87% Source: Page 21

FIND-008 Insecure File Upload — Document Submission [High]

CVSS: 8.5 **Asset:** https://egovportal.gov.in/citizen/upload **CVE:** CVE-2023-50164 **CWE:** CWE-434

Recommendation: Validate file type server-side using magic bytes. Store uploads outside web root. Scan for malware.

Confidence: 90% Source: Page 28

5.3 OWASP Top 10 2021 Coverage — Web Application

The system automatically populates the webapp_specific.owasp_top10_2021_coverage array in the standardised output:

OWASP Category	Findings Count	Status
A01:2021 – Broken Access Control	3	FAIL
A02:2021 – Cryptographic Failures	1	FAIL
A03:2021 – Injection	3	FAIL
A04:2021 – Insecure Design	0	PASS
A05:2021 – Security Misconfiguration	2	FAIL
A06:2021 – Vulnerable Components	1	FAIL
A07:2021 – Auth & Session Failures	2	FAIL
A08:2021 – Software & Data Integrity	0	PASS

A09:2021 – Logging & Monitoring	1	PARTIAL
A10:2021 – SSRF	0	PASS

Web Application Output Summary	
Submission Reference	SRN-2026-WAST-C9D1E5G3
Total Findings	11
Critical	2 | High: 5 | Medium: 3 | Low: 1
OWASP Categories Tested	10 / 10 (full coverage)
OWASP Categories Failed	7 of 10
Evidence Linked	14 of 11 findings have at least 1 evidence item
Avg. Confidence	90%
Processing Time	~81 seconds

6. Domain Demo 4 — IoT Security Audit

6.1 Sample Input Report — Description

Sample Input File
Filename: iot_security_smartcity_sensors_sep2026.pdf | Pages: 38 | Format: PDF (Partially Scanned — 6 pages scanned) | Device Count: 24 IoT devices | Agency: IoT Defence Labs Pvt. Ltd. | Audit Type: IoT Security Audit (Smart City Infrastructure)

This IoT audit report presents unique challenges: 6 of 38 pages are scanned (containing physical device photographs and PCB close-ups showing UART/JTAG interfaces). Section headings include 'Device Assessment', 'Firmware Analysis', 'Communication Security', 'Physical Interface Findings', and 'OWASP IoT Compliance'. The system handles all of these.

6.2 Key Processing Stages — IoT Audit

STAGE 05 **OCR Processing — IoT Report (Partially Scanned)**

OCR Stage Output	
OCR Applied?	True — applied to 6 scanned pages (Pages 15, 16, 17, 24, 25, 31)
OCR Engine	Tesseract v5 LSTM + OpenCV preprocessing
Preprocessing Applied	Greyscale → Deskew (avg 1.2° correction) → Adaptive Threshold → Denoise
Avg. Confidence (scanned pages)	0.79 (Page 15 — low-quality PCB photo: 0.61)
Page 15 Flag	Low confidence (0.61) — flagged for human review
Native Pages	32 pages extracted natively (confidence 1.0)

STAGE 09 **Rule-Based Parsing — IoT Report**

Rule Name	Matches	Sample Values
IOT_PROTOCOL	14	MQTT (x5), Zigbee (x3), CoAP (x2), BLE (x2), Modbus (x1), LoRaWAN (x1)

FIRMWARE_VERSION	8	firmware v1.2.3, v2.0.1, v0.9.7-beta (3 devices on outdated firmware)
IP_V4	24	10.0.1.1 – 10.0.1.24 (one IP per IoT device)
MAC_ADDRESS	24	24 unique MAC addresses — OUI mapped to 3 manufacturers
CVE_ID	6	CVE-2023-26035, CVE-2024-0778, CVE-2023-49897 ...
MQTT_TOPIC	7	'/sensors/temp', '/devices/status', '/gateway/cmd' — all unencrypted

STAGE 10 **AI Entity Extraction — IoT Findings**

Four representative IoT findings with domain-specific fields populated:

FIND-001 Default Credentials on MQTT Broker [Critical]

CVSS: 9.1 **Asset:** IoT Gateway — 10.0.1.1 (MQTT Port 1883) **CVE:** — **CWE:** CWE-1392

Recommendation: Change MQTT broker credentials immediately. Enforce client certificate authentication. Enable TLS.

Confidence: 93% Source: Page 9

FIND-002 UART Debug Interface Exposed on Smart Meter PCB [High]

CVSS: 8.4 **Asset:** Smart Energy Meter — Device ID SM-07 **CVE:** CVE-2023-26035 **CWE:** CWE-1233

Recommendation: Remove or disable UART pads in production firmware. Implement secure boot to prevent root shell access.

Confidence: 88% Source: Page 16

FIND-004 Unencrypted MQTT Communication — Sensor Data [High]

CVSS: 8.1 **Asset:** All 8 temperature sensors — 10.0.1.5–10.0.1.12 **CVE:** — **CWE:** CWE-319

Recommendation: Enable MQTT TLS (port 8883). Validate broker certificates. Rotate all symmetric keys.

Confidence: 91% Source: Page 21

FIND-006 Hardcoded Admin Credentials in Firmware v1.2.3 [Critical]

CVSS: 9.8 **Asset:** IoT Camera Devices — 10.0.1.18–10.0.1.21 **CVE:** CVE-2024-0778 **CWE:** CWE-798

Recommendation: Issue emergency firmware update. Revoke hardcoded credentials. Implement OTA update with signed firmware.

Confidence: 94% Source: Page 27

6.3 IoT-Specific Output Fields

```
"iot_specific": {
  "device_inventory": [
    { "device_id": "GW-01", "device_type": "IoT Gateway",
      "manufacturer": "Huawei", "firmware_version": "v2.0.1",
      "ip_address": "10.0.1.1", "communication_protocols": ["MQTT","Zigbee"],
      "criticality": "Critical" },
    { "device_id": "SM-07", "device_type": "Smart Energy Meter",
      "manufacturer": "Genus Power", "firmware_version": "v0.9.7-beta",
      "ip_address": "10.0.1.7", "communication_protocols": ["Modbus","BLE"],
      "criticality": "High" }
  ],
  "communication_security_summary": {
    "mqtt_tls_enabled": false,
    "unencrypted_protocols_found": ["MQTT", "CoAP (no DTLS)"],
    "certificate_validation_issues": true
  },
  "firmware_analysis_summary": {
    "firmware_extracted": true,
    "hardcoded_credentials_found": true,
    "debug_symbols_present": true,
    "insecure_functions_found": ["strcpy", "gets", "sprintf"]
  },
  "physical_security_summary": {
    "jtag_exposed": false,
    "uart_exposed": true,
    "spi_exposed": false,
    "tamper_protection_present": false
  }
}
```


IoT Audit Output Summary	
Submission Reference	SRN-2026-IOT-D4F8B2C7
Total Devices	24 IoT devices in scope
Total Findings	14
Critical	3 | High: 6 | Medium: 4 | Low: 1
OWASP IoT Top 10 Coverage	9 of 10 categories tested
Firmware Issues	3 devices on outdated firmware; hardcoded credentials found
Physical Security	UART exposed on 4 devices; no tamper protection on any device
OCR Flagged Pages	Page 15 (conf: 0.61) — flagged for manual review
Avg. Confidence	87%
Processing Time	~95 seconds (includes OCR for 6 scanned pages)

7. Source Traceability — Demonstration

7.1 What Source Traceability Means

Every single field extracted by the AI or rule parser is tagged with the exact location in the original uploaded document where that value was found. A reviewer can click any field in the Output Viewer and immediately see: which file, which page, which paragraph, and the exact raw text snippet — allowing them to verify the AI extraction against the original.

7.2 Traceability in Action — VAPT Example

The following shows the source_traceability object attached to Finding FIND-001 (SQL Injection) from the VAPT demo:

```
"source_traceability": {
  "source_file":      "vapt_report_agency_abc_sep2026.pdf",
  "page_number":      9,
  "section_heading":  "Observations",
  "paragraph_number": 2,
  "table_id":         "TABLE-02",
  "row_number":       1,
  "raw_text_snippet": "Finding 1: SQL Injection in Login API (Critical, CVSS 9.8)"
                      "The /api/v1/login endpoint accepts unsanitised user input..."
}
```

7.3 Traceability Coverage Across All Four Demos

Domain	Total Findings	Findings with Full Traceability	Findings with Partial Traceability	Not Traced
VAPT	12	11 (92%)	1 (8%)	0
Network Infrastructure	15	14 (93%)	1 (7%)	0
Web Application	11	10 (91%)	1 (9%)	0
IoT Security	14	12 (86%)	2 (14%)	0

Key Observation

Partial traceability occurs when the AI paraphrased or merged content from multiple paragraphs. In these cases, the tracer links to the most probable source page but cannot pin an exact paragraph. Such findings are shown with amber (Medium) confidence in the Output Viewer.

8. AI Synonym Mapping — Demonstration

8.1 The Problem Illustrated

The table below shows actual section headings collected from the four sample reports used in the demos. All of these varied headings were successfully mapped to their correct standard section by the system's three-tier classification approach:

Agency / Report	Original Heading Used	Standard Section Name	Method	Confidence
VAPT — Agency ABC	Observations	Vulnerability Findings	Synonym Rule	92%

VAPT — Agency ABC	Corrective Measures	Recommendations	Synonym Rule	100%
VAPT — Agency ABC	Closing Remarks	Conclusion	Synonym Rule	91%
Network — SecureNet	Network Assessment Findings	Vulnerability Findings	ModernBERT	84%
Network — SecureNet	Configuration Issues	Vulnerability Findings	ModernBERT	81%
Network — SecureNet	Hardening Recommendations	Recommendations	Synonym Rule	88%
Web App — AppSec India	Vulnerability Details	Vulnerability Findings	Synonym Rule	94%
Web App — AppSec India	Risk Classification	Vulnerability Findings	BGE-M3 Similarity	76%
Web App — AppSec India	Fixes and Mitigations	Recommendations	Synonym Rule	97%
IoT — IoT Defence Labs	Device Assessment	Assets Covered	BGE-M3 Similarity	73%
IoT — IoT Defence Labs	Firmware Analysis	Vulnerability Findings	ModernBERT	86%
IoT — IoT Defence Labs	Physical Interface Findings	Vulnerability Findings	ModernBERT	82%

8.2 Synonym Mapping Statistics — Across All Four Demos

Mapping Method	Headings Mapped	Success Rate	Avg. Confidence
Synonym Rule (Exact Match)	18	100%	99%
Synonym Rule (Partial Match)	9	100%	91%
ModernBERT Classifier	11	91% (1 low-conf corrected by BGE-M3)	85%
BGE-M3 Similarity	6	83% (1 marked 'Other' — insufficient context)	75%
Unclassified (Other)	1	—	< 50%

9. Severity Normalisation — Demonstration

9.1 Raw Severity Labels Found Across All Four Reports

The four sample reports used a total of 19 different raw severity expressions. The SetFit-based severity normaliser mapped all 19 to the 5-class standard:

Raw Label Found in Report	Count	Domain	→ Normalised Label	Method
Critical	5	All	Critical	Synonym (Exact)
P0	2	VAPT	Critical	SetFit
Emergency	1	Network	Critical	SetFit
High	11	All	High	Synonym (Exact)
Major	3	VAPT, IoT	High	SetFit
Important	2	Web	High	SetFit

Medium	8	All	Medium	Synonym (Exact)
Moderate	4	VAPT, Web	Medium	Synonym
P2	2	Network	Medium	SetFit
Orange	1	Web	Medium	SetFit
Low	5	All	Low	Synonym (Exact)
Minor	3	VAPT, IoT	Low	SetFit
Low Risk	2	Network	Low	Synonym
Informational	3	All	Informational	Synonym (Exact)
Info	2	VAPT, Web	Informational	Synonym
Advisory	1	Network	Informational	SetFit
Best Practice	2	IoT	Informational	SetFit
N/A	1	Network	Informational	CVSS Score
P-0 (Urgent)	1	Web	Critical	SetFit

Normalisation Accuracy
All 19 severity expression types were correctly normalised across all four demos. SetFit handled 8 of the 19 non-standard labels with 100% accuracy — all verified against the original reports.

10. Final Standardised Output — All Formats

10.1 Output Formats Generated

For each processed report, the system generates four output artifacts, all saved to the outputs/SRN-XXXX/ directory:

Format	Filename	Purpose	Audience
JSON	structured_report.json	Machine-readable standard output — primary format for integration, database ingestion, dashboard feeds	System integrators, developers, automated analytics
XML	structured_report.xml	Alternative machine-readable format — for legacy systems or XML-based integration pipelines	Legacy system integrators
PDF Summary	summary_report.pdf	Human-readable summary — executive findings, severity chart, top recommendations, traceability stats	Management, evaluators, stakeholders
ZIP Package	full_package.zip	All of the above + original file hash record + evidence files + processing log	Archiving, audit trail, compliance

10.2 JSON Output Structure — Top-Level Preview (VAPT Demo)


```
{
  "report_metadata":       { "srn": "SRN-2026-VAPT-A3F2C1D4", "schema_version": "1.0.0",
... },
  "audit_type":            { "primary_type": "VAPT", "testing_approach": "Grey-Box" },
  "audited_organisation":  { "name": "XYZ Government Agency", "sector": "Government" },
```

```
  "auditing_agency":          { "name": "XYZ Cyber Security Pvt. Ltd.", "empanelment_id":
"CERT-IN-EA-123" },
  "audit_scope":          { "description": "...", "in_scope_systems": ["192.168.1.100", ...] },
  "tools_used":          [ { "tool_name": "Burp Suite", "tool_category": "Proxy" }, ... ],
  "vulnerability_findings": [ { "finding_id": "FIND-001", "severity": "Critical", ... }, ... ],
  "executive_summary":     { "total_findings": 12, "overall_risk_rating": "Critical" },
  "approval_status":       { "status": "Pending" }
}
```


10.3 Comparative Output Across All Four Domains

Metric	VAPT	Network	Web App	IoT
SRN	SRN-2026-VAPT-A3F2C1D4	SRN-2026-NET-B7E4A2F1	SRN-2026-WAST-C9D1E5G3	SRN-2026-IOT-D4F8B2C7
Input Format	PDF (28 pages)	DOCX (34 pages)	PDF (42 pages)	PDF (38 pages, 6 scanned)
OCR Applied?	No	No	No	Yes (6 pages)
Total Findings	12	15	11	14
Critical	2	3	2	3
High	5	8	5	6
CVEs Mapped	5	8	4	6
Evidence Items	10	4	19	8
Avg. Confidence	91%	89%	90%	87%
Fully Traced	92%	93%	91%	86%
Processing Time	~68 sec	~72 sec	~81 sec	~95 sec
Output Size (JSON)	~48 KB	~61 KB	~52 KB	~74 KB

11. Evaluation and Performance Metrics

11.1 Section Detection Accuracy

Across all four domain demos, 45 unique section headings were detected. Mapping accuracy was assessed by manually verifying each mapped section against the original report:

Metric	Value
Total headings detected	45
Correctly mapped to standard section	43 (95.6%)
Incorrectly mapped	1 (2.2%) — 'Network Assessment Findings' mapped to 'Technical Summary' instead of 'Vulnerability Findings'; corrected by reviewer
Unclassified (Other)	1 (2.2%) — 'Consultant Bio' section — no equivalent in standard schema
Synonym Rule (Exact + Partial)	27 headings (60%)
ModernBERT Classifier	11 headings (24%)
BGE-M3 Similarity	6 headings (13%)
Unclassified	1 heading (2%)

11.2 Entity Extraction Accuracy

Entity Type	Total Expected	Extracted	Accuracy	Notes
CVE Identifiers	23	22	95.6%	1 CVE missed — appeared only in a scanned screenshot caption
CVSS Scores	23	21	91.3%	2 CVSS scores not explicitly stated in report — inferred as blank
Severity Labels	52	52	100%	All raw labels extracted; all 19 types normalised correctly
Affected IP Addresses	38	36	94.7%	2 IPs in a low-quality scanned table (OCR conf 0.61) — missed
URLs / Endpoints	31	31	100%	All URLs successfully extracted by regex rule
Tool Names	14	13	92.8%	1 tool ('Greenbone Security Manager') not in known-tools list
Finding Records	52	50	96.2%	2 findings merged by LLM into 1; identified during review

11.3 Processing Performance

Component	Avg. Time (text PDF)	Avg. Time (scanned PDF)	Avg. Time (DOCX)
File Validation + Hashing	< 1 sec	< 1 sec	< 1 sec
Malware Scan (ClamAV)	3–8 sec	3–8 sec	3–8 sec
OCR (Tesseract + OpenCV)	N/A	15–45 sec	N/A
Text + Table Extraction	2–5 sec	1–3 sec	3–8 sec
AI Section Detection	8–15 sec	8–15 sec	8–15 sec
Rule-Based Parsing	< 1 sec	< 1 sec	< 1 sec
AI Entity Extraction (Mistral-7B)	45–90 sec	45–90 sec	45–90 sec
Severity Normalisation (SetFit)	< 2 sec	< 2 sec	< 2 sec
Traceability Tagging	2–5 sec	2–5 sec	2–5 sec
Output Generation	2–4 sec	2–4 sec	2–4 sec
TOTAL (CPU mode)	65–120 sec	80–160 sec	65–120 sec
TOTAL (GPU mode, 8GB VRAM)	20–40 sec	35–60 sec	20–40 sec

12. Appendices

Appendix A — Demo Quick Reference

Demo	Input File	SRN	Findings	Risk	Time
VAPT	vapt_report_agency_abc_sep2026.pdf	SRN-2026-VAPT-A3F2C1D4	12	Critical	~68 sec

Network	network_audit_ministry_oct2026.docx	SRN-2026-NET-B7E4A2F1	15	Critical	~72 sec
Web App	webapp_security_egovportal_aug2026.pdf	SRN-2026-WAST-C9D1E5G3	11	Critical	~81 sec
IoT	iot_security_smartcity_sensors_sep2026.pdf	SRN-2026-IOT-D4F8B2C7	14	Critical	~95 sec

Appendix B — Confidence Score Interpretation

Confidence Range	Indicator	Meaning	Action Required
85% – 100%	High (Green)	Extracted from clearly structured, well-formatted source text	No review required — may approve directly
60% – 84%	Medium (Amber)	AI-assisted inference used; some ambiguity in source	Reviewer should verify against original document
< 60%	Low (Red)	Significant ambiguity or poor source quality	Mandatory review and correction before approval
Not Found	Grey	Field not present in source document	Check original; may be genuinely missing

Appendix C — Domain-Specific Field Population Summary

Domain	Domain Object	Key Fields Populated
VAPT	domain_specific_fields.vapt	exploit_path, privilege_escalation_possible, patch_available, lateral_movement_possible
Network	domain_specific_fields.network	protocol, port_number, service_name, firewall_rule_issue, encryption_issue, network_segment
Web Application	domain_specific_fields.web_application	owasp_category, http_method, vulnerable_url, authentication_bypass, input_validation_issue
IoT	domain_specific_fields.iot	device_type, communication_protocol, owasp_iot_category, default_credentials_found, firmware_extraction_possible, debug_interface_exposed